



Nutrition and Disease

Pilot Randomized Controlled Trial of Krill Oil Supplementation for Chronic Musculoskeletal Pain in Older Adults

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ABSTRACT

Background: Chronic musculoskeletal (MSK) pain is highly prevalent in older adults and the leading cause of disability. Long-chain omega-3 (ω -3) polyunsaturated fatty acids have anti-inflammatory and analgesic properties, yet intake and circulating levels are low in older US adults. Krill oil, a phospholipid-bound source of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) that also contains astaxanthin and choline, may represent a feasible nutrition-based intervention for chronic MSK pain.

Objectives: To evaluate the feasibility and acceptability of krill oil supplementation in older adults with chronic MSK pain and to characterize changes in ω -3 status and exploratory pain outcomes.

Methods: A 12-wk, randomized, double-blind, placebo-controlled pilot trial in 40 community-dwelling adults aged ≥ 60 y with moderate chronic MSK pain. Participants were stratified by sex and randomly assigned 1:1 to krill oil (4 g/d; 1288 mg EPA+DHA) or placebo. Primary outcomes were feasibility metrics (recruitment, retention, adherence, and acceptability). Secondary outcomes included biomarkers of ω -3 status and inflammation. Exploratory outcomes included pain intensity, pain interference, pain distribution, and global improvement.

Results: Recruitment targets were met within 8 mo; retention was 95%, adherence exceeded prespecified thresholds ($>70\%$) in both groups, and overall acceptability was rated 7.7 ± 2.3 of 10. Krill oil supplementation increased the ω -3 index from 4.3% to 7.4%, approaching levels consistent with lower cardiometabolic risk ($\geq 8\%$), whereas the placebo group showed no meaningful change. Preliminary analyses showed trends favoring krill oil for reduced pain intensity ($\Delta KO - \Delta P = -1.05$), fewer pain sites (rate ratio: 0.59; 95% confidence interval: 0.41, 0.84), and global improvement (adjusted odds ratio: 4.35; 95% confidence interval: 0.97, 19.55). No serious adverse events occurred.

Conclusions: Krill oil supplementation was feasible and acceptable among older US adults with chronic MSK pain. These findings support a fully powered randomized controlled trial to evaluate krill oil as a nutritional strategy for pain management in aging populations. This trial was registered at clinicaltrials.gov as NCT06580912.

Keywords: chronic pain, ω -3 fatty acids, dietary supplements, feasibility studies

Introduction

Chronic musculoskeletal (MSK) pain is disproportionately prevalent among older adults [1] and is the leading cause of disability globally [2]. Pain management in this population remains particularly challenging and often ineffective,

complicated by multimorbidity, age-related physiological changes, and concerns related to polypharmacy [3]. Opioid analgesics used to manage moderate-to-severe pain carry the risk of adverse effects such as dependence and tolerance, cognitive impairments, and drug-drug interactions in older adults, highlighting the need for non-pharmacological

Abbreviations: MSK, musculoskeletal; PEG, pain, enjoyment of life, and general activity; PGIC, patient global impression of change; WOMAC, Western Ontario and McMaster Universities osteoarthritis index.

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approaches [3–5]. Nutrition-based interventions are particularly appealing as they are non-invasive, cost-effective, highly individualized, and generally low risk. Despite the growing burden of chronic pain in aging populations [6], safe and scalable nutrition strategies that target MSK pain remain underexplored.

Most older adults in the United States do not meet the recommended intakes of omega (ω)-3 long-chain PUFAs (LC-PUFAs), and <2% exhibit circulating levels associated with a reduced risk of cardiovascular disease and mortality [7–9]. Due to their well-recognized anti-inflammatory properties [10], ω -3 LC-PUFAs, particularly EPA and DHA, have been increasingly studied for their potential analgesic effects. Mechanistically, ω -3 LC-PUFAs may modulate pain both peripherally and centrally by promoting specialized pro-resolving mediators, dampening neuronal excitability, and supporting nerve repair and neuro-immune homeostasis [11]. Additionally, ω -3 LC-PUFAs may support MSK pain specifically by attenuating catabolic processes and the progression of osteoarthritis (OA) [12,13]. Meta-analyses of ω -3 supplementation trials have found moderate, clinically meaningful reductions for inflammatory pain [11,14,15], whereas evidence for OA-related pain remains limited and inconclusive [11]. As suggested by Xie et al. [11], the lack of statistically significant effects observed in some OA trials may reflect the complex and multifactorial nature of OA-related pain, including low-grade inflammation, mechanical drivers of pain, and the use of multidimensional outcome measures [11]. Moreover, relatively few trials have specifically examined OA populations.

Krill oil (*Euphausia superba*) is a marine source of ω -3 LC-PUFAs in which EPA and DHA are predominantly bound to phospholipids rather than triglycerides or ethyl esters. This structural difference is thought to enhance bioavailability and tissue uptake [16]. On the contrary, the absolute EPA and DHA content per gram in krill oil is often lower than that of concentrated fish oils [17,18], which may offset its benefits by the need for higher volumes to achieve the desired intake of EPA and DHA, potentially compromising tolerability and acceptability, especially in older adults. Beyond ω -3 fatty acids, krill oil contains additional bioactive compounds that may be beneficial for pain. Krill is naturally rich in astaxanthin, a xanthophyll carotenoid with potent anti-inflammatory/antioxidant, geroprotective, and analgesic properties [19,20]. In addition, krill oil provides choline as phosphatidylcholine, an essential nutrient involved in neurotransmission, cell signaling, lipid metabolism, and homocysteine regulation. As a precursor to acetylcholine, choline may influence cholinergic pathways involved in pain modulation [21] and plays an important role in skeletal muscle metabolism and musculoskeletal health [22]. Yet, choline is poorly consumed by older adults in the United States [23,24]. Although krill oil supplementation has demonstrated favorable effects on knee OA [25], older adults have been underrepresented in these trials. As such, the feasibility, acceptability, and potential effects of krill oil on chronic MSK pain in older adults remain unknown.

The Krill Oil for Pain in Elders (KOPE) pilot study was a randomized, double-blind, placebo-controlled trial designed to evaluate the feasibility of conducting a larger efficacy trial in older adults with chronic MSK pain. Consistent with the CONSORT extension for pilot and feasibility trials, the primary objectives of the KOPE pilot were to assess feasibility metrics,

including recruitment, retention, adherence, acceptability, and data completeness, to inform the design of a future fully powered randomized controlled trial. Our secondary objectives were to characterize changes in ω -3 status and inflammation, and we performed preliminary analyses of pain outcomes.

Methods

Trial design

This pilot trial used a randomized, double-blind, placebo-controlled, parallel-group design conducted over a 12-wk intervention period. Participants completed 3 in-person study visits: a baseline visit prior to randomization (week 0), a mid-intervention follow-up visit (week 6), and a final assessment at the end of the intervention period (week 12). The trial was designed and reported in accordance with the CONSORT extension for randomized pilot and feasibility trials (see [Supplementary Checklist](#)) [26]. All research was conducted at the University of Florida. The study was approved by the University of Florida Institutional Review Board (#202400581) and conducted under an active US Food and Drug Administration Investigational New Drug application. All participants provided written informed consent prior to participation. The trial is registered with [clinicaltrials.gov](#) (NCT06580912); full details are available at <https://clinicaltrials.gov/study/NCT06580912>.

Participants

A total of 40 community-dwelling older adults were recruited from Gainesville, FL, and neighboring areas primarily through postcard mailings and social media advertisements. Participants were first screened for preliminary eligibility by telephone and subsequently invited for an in-person screening visit, during which written informed consent and eligibility assessments were completed. Eligible participants completed the baseline assessment within 30 d of the screening visit prior to randomization. If >30 d elapsed between screening and baseline, eligibility assessments were repeated to confirm continued eligibility.

Eligible participants were males and females aged ≥ 60 y who reported moderate pain (≥ 4 on a 0–10 scale) [27] in the hips, knees, or lower back on most days for the past 3 mo and demonstrated moderate mobility limitations. Exclusion criteria were designed to minimize risks and reduce confounding. These included: bleeding or coagulation disorders; use of anticoagulants, full-dose aspirin, opioids, high-dose nonsteroidal anti-inflammatory drugs, selective serotonin reuptake inhibitors, and complementary medicines/supplements that may pose a risk of food-drug interaction (e.g., St. John's Wort) or may affect study results; ω -3 supplementation (<3 mo) or >2 servings per week of fatty fish intake; seafood allergy; clinically significant or uncontrolled medical conditions; substance use history; severe mobility impairment; or planned surgery during the study period.

Randomization and blinding

Participants were stratified by sex and randomly assigned in a 1:1 ratio to receive either krill oil or a placebo. Randomization

was performed by (YC-A) using a computer-generated allocation sequence with block randomization using varying block sizes (2, 4, and 6) to ensure balance between groups. Allocation concealment was maintained by using identical-appearing study capsules and bottles. Except for (YC-A), participants, investigators, and study staff who interacted with participants were blinded to group assignment throughout the trial. (YC-A) was not involved in patient recruitment, enrollment, or clinical assessments.

Intervention

Participants assigned to the treatment group received 4 g/d of krill oil (Superba Boost, Aker BioMarine Antarctic US LLC) in the form of 4 1000 mg capsules, providing a total of 1288 mg/d EPA+DHA, 0.45 mg astaxanthin, and 320 mg choline daily. Placebo capsules matched the krill oil capsules in appearance, odor, and taste, containing 1000 mg of mixed vegetable oil, comprised olive oil, corn oil, palm oil, and medium-chain triglycerides in a ratio of 4:4:3:2 (~31% SFAs, 46% MUFAs, and 22% PUFAs, with no EPA or DHA). This mixture resembles the fatty acid composition of a typical Western diet [28,29]. Within the context of the whole diet, the small amount of mixed fats consumed in the placebo is not expected to have any independent effects. Participants were instructed to take the assigned capsules once daily, with a meal, for 12 wk and to maintain their usual diet and physical activity patterns throughout the study period. Tolerance was assessed weekly for the first 3 wk, and dosage and timing were adjusted as needed. If participants reported digestive discomfort or difficulty tolerating 4 capsules per day, they were advised to temporarily reduce intake to 1 capsule and gradually increase by 1 capsule daily as tolerated until the full daily dose was reached. If taking all capsules at once was not well tolerated, the dose could also be divided across the day (e.g., 2 capsules in the morning and 2 in the evening).

Data collection

Study data were collected and managed using REDCap (Research Electronic Data Capture) hosted at the University of Florida [30,31].

- Feasibility outcomes (primary): feasibility was determined by meeting pre-selected criteria, which included achieving $\geq 80\%$ of the recruitment goal within 12 mo, maintaining $< 20\%$ attrition, and achieving $> 70\%$ adherence to the intervention. The Medication Acceptability Questionnaire (MAQ) was used to assess perceptions of convenience, taste and texture, appearance and smell, side effects, and overall acceptance [32].
- Biomarker outcomes (secondary): fasted whole-blood samples were obtained at each study visit. Changes in blood fatty acid profile were assessed from dried blood spots using gas chromatography (OmegaQuant Analytics) [33]. The ω -3 index is calculated as the percentage of EPA and DHA in erythrocytes, and is considered a clinically relevant biomarker of ω -3 status that is highly responsive to dietary intake [34,35]. The circulating ω -6: ω -3 ratio was included for contextualization with prior studies, although it is no longer considered a preferred indicator of ω -3 status.

High-sensitivity C-reactive protein (hs-CRP) was measured at every visit as an inflammatory biomarker and for safety monitoring. Hs-CRP concentrations were quantified in serum using a high-sensitivity immunoturbidimetric assay and reported in milligrams per liter.

- Clinical outcomes (exploratory): due to small sample sizes, pilot studies cannot adequately provide evidence of safety, tolerability, or effect sizes. Here, we present preliminary analyses focused on the primary endpoints planned for a future, fully powered trial: pain intensity and interference, and physical function.

Average pain intensity (past week) was assessed using a single item from the Graded Chronic Pain Scale-Revised (0–10 scale), which includes the 3-item Pain, Enjoyment of Life, and General Activity (PEG) scale used as a summary measure of pain intensity and interference [36]. The Western Ontario and McMaster Universities OA Index (WOMAC) was used to measure joint pain, stiffness, and physical function; all scores were normalized to a 0–100 scale [37]. The overall perceived treatment response was assessed using the Patient Global Impression of Change (PGIC) [38], dichotomized as “improved” compared with “not improved.” Pain distribution was quantified as the total number of painful body sites using the Michigan Body Map (range: 0–32) [39]. Neuropathic pain symptoms were assessed with the painDETECT [40]. Self-reported physical function was assessed with the Pepper Assessment Tool for Disability, which evaluates mobility, transferring, upper extremity function, activities of daily living, and instrumental activities of daily living [41,42]. Performance-based physical function was assessed using the short physical performance battery (SPPB), which includes tests of balance, gait speed, and chair stands, with a total score ranging from 0 to 12 [43,44]. Handgrip strength was measured using a handheld dynamometer as the highest value derived from the mean of 2 trials performed on each hand [45]. Functional capacity and endurance were assessed using the 6-min walk test [46].

- Additional measures: sociodemographic characteristics (age, sex, race and ethnicity, as well as education) were self-reported. Height and weight were measured using standardized procedures and used to calculate the BMI (in kg/m²). Dietary intake was assessed using the Diet History Questionnaire III to measure total energy intake, overall diet quality using the Healthy Eating Index (HEI), and dietary fat quality via the HEI fatty acids component. HEI scores range from 0 to 100, with higher scores indicating greater adherence to the *Dietary Guidelines for Americans*.

Sample size

This pilot study was designed to guide the design and implementation of large-scale trials. We planned to enroll 40 older adults within a 12-mo period. Randomization was stratified by sex to ensure balanced representation across treatment arms (10 males and 10 females per arm). Allowing an anticipated attrition rate of ~20%, we expected ~32 participants (~16 per group) to complete the intervention. This sample size is appropriate for assessing feasibility outcomes and is consistent with methodological recommendations for pilot randomized trials, particularly when anticipated effects are small to moderate [47].

Statistical analysis

Analyses of feasibility outcomes were descriptive in nature and summarized using proportions and means with SDs. Changes in biomarkers and clinical outcomes were evaluated using mixed-effects models with fixed effects for treatment group, time (modeled as a categorical variable), and the group-by-time interaction, a random intercept for each participant, and an independent covariance structure. All models were adjusted for age, sex, and BMI. For continuous outcomes, treatment effects are reported as adjusted between-group differences in change from baseline ($\Delta KO - \Delta P$) with 95% confidence intervals (CIs). Hs-CRP was log-transformed prior to analysis due to right-skewed distributions. Total pain sites, a count outcome, were analyzed using negative binomial mixed-effects models and are reported as rate ratios with 95% CIs. The PGIC was analyzed as a binary outcome using logistic regression, with results reported as odds ratios and 95% CIs. Standardized effect sizes (Cohen's *d*) were also reported for continuous outcomes to aid interpretation of effect magnitude, calculated using the residual SD from the mixed-effects models. Effect sizes were interpreted using conventional thresholds, with values of ~ 0.2 , ~ 0.5 , and ~ 0.8 representing small, medium, and large effects, respectively. Model assumptions were evaluated through standard diagnostic procedures, including inspection of residual distributions and assessment of homoscedasticity. All analyses followed the intention-to-treat principle. Statistical analyses were conducted using SAS version 9.4 (SAS Institute).

Results

A participant flowchart is presented in Figure 1. Recruitment began in January 2025 and continued through September 2025. During this period, 334 individuals were phone-screened for initial eligibility. Of these, 63 participants (18.9%) met preliminary criteria and were invited to complete an in-person

screening visit. A total of 23 individuals were excluded during screening. The most common reasons for exclusion were SPPB scores outside the eligible range (>10 , $n = 11$; <4 , $n = 1$) and a history of clinically significant conditions ($n = 6$). One individual declined participation during the informed consent process.

A total of 40 participants were subsequently randomly assigned to receive either krill oil ($n = 20$) or placebo ($n = 20$) in a sex-stratified manner (10 males and 10 females per group). All randomly assigned participants received their allocated intervention and were included in the intention-to-treat analyses.

Participant characteristics

Baseline characteristics were generally similar between groups (Table 1). Participants had a mean age of 70.1 y (SD = 5.7), were predominantly non-Hispanic White (72.5%), and had a mean BMI of 30.8 (SD = 6.7). The proportion of non-Hispanic White participants was slightly higher in the placebo group. Diet quality was similar between groups, although mean total energy intake was higher in the placebo group.

Participants reported a high pain burden, with a mean of 7.8 (SD = 4.2) pain sites per participant (range: 2–21). The most common pain sites were the lower back (75.0%), hips (65.0%), and knees (62.5%). The lower back was also the most frequently reported worst pain site (30.0%) and second-worst pain site (12.5%), experienced for durations of 10 (IQR: 3, 16) and 5 (3, 25) y, respectively. Participants also reported moderate to high pain intensity and interference, with a mean PEG score of 5.5 (SD = 2.3) and a WOMAC score of 40.1 (SD = 13.5). Nearly half (47.5%) of participants were classified as having possible or likely neuropathic pain, with a mean PainDETECT score of 12.7 (SD = 6.8).

Performance-based physical function tests showed moderate impairments, with a mean SPPB score of 8.5 (SD = 1.4), 6-min walk test distance of 368.0 m (SD = 85.1), and handgrip strength of 28.9 kg (SD = 9.7).

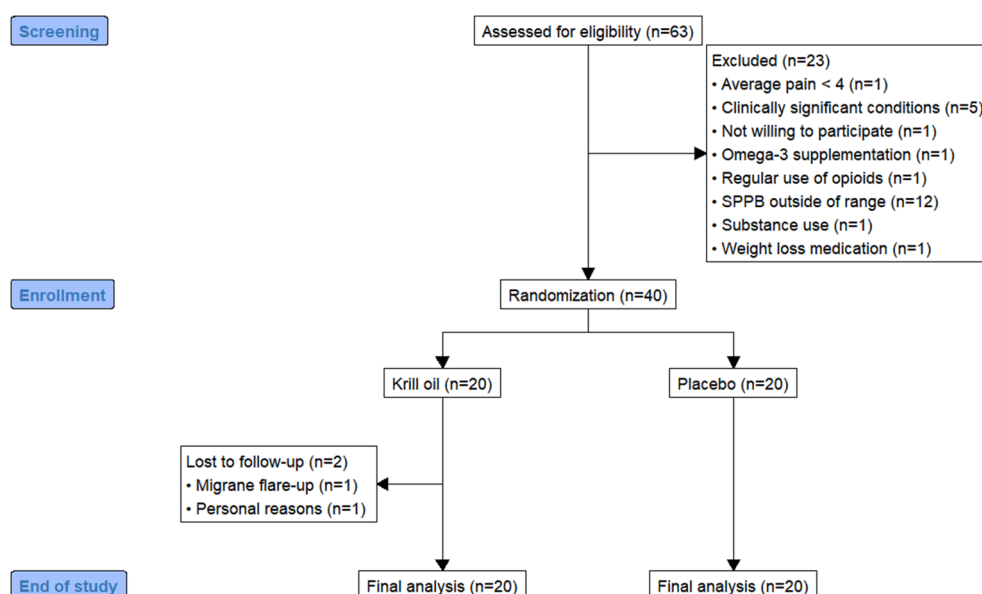


FIGURE 1. Krill Oil for Pain in Elders (KOPE) pilot participant flowchart. SPPB, Short Physical Performance Battery.

TABLE 1

Baseline characteristics of participants in the Krill Oil for Pain in Elders (KOPE) pilot trial

Characteristic	Krill oil (n = 20)	Placebo (n = 20)	Standardized difference ¹
Age, y	69.4 ± 5.5	70.7 ± 5.9	0.23
Sex, female	10 (50)	10 (50)	0.00
Race and ethnicity			
Non-Hispanic White	13 (65.0)	16 (80.0)	0.49
Non-Hispanic Black	1 (5.0)	2 (10.0)	
Hispanic	4 (20.0)	2 (10.0)	
Asian	2 (10.0)	0 (0)	
Education			
High school or less	8 (40.0)	3 (15.0)	0.48
Some college	2 (10.0)	5 (25.0)	
Bachelor's or higher	10 (50.0)	12 (60.0)	
Diet			
Total energy intake (kcal/d)	1326 ± 423	1652 ± 860	0.49
HEI ²	62.7 ± 10.4	63.8 ± 7.9	0.12
HEI fatty acids (0–10) ³	4.6 ± 3.2	4.7 ± 3.0	0.03
BMI (kg/m ²)	29.9 ± 7.5	31.8 ± 5.9	0.28
Physical performance			
SPPB total score (0–12)	9.0 ± 1.1	8.1 ± 1.6	0.66
6-min walk distance (m)	384.1 ± 63.8	352.0 ± 101.2	0.38
Handgrip strength (kg)	30.2 ± 8.4	27.7 ± 10.9	0.26

Values are shown as mean ± SD or n (%).

Abbreviations: HEI, Healthy Eating Index; SPPB, Short Physical Performance Battery.

¹ Standardized differences were calculated as the difference in means or proportions divided by the pooled SD. Absolute values are presented.

² HEI scores range from 0 to 100, with higher scores indicating greater adherence to the *Dietary Guidelines for Americans*.

³ HEI fatty acids component reflects dietary fat quality and is calculated as the ratio of (PUFAs+MUFAs)/SFAs, expressed as a percentage (%) of total energy. Scores range from 0 to 10 points, with 0 points for a ratio ≤1.2 and 10 points assigned for a ratio ≥2.5; intermediate values are scored proportionally.

Feasibility outcomes

Recruitment took place over ~8 mo, with an average of 5 participants enrolled per month. Notably, enrollment differed by sex. The target enrollment for females (n = 20) was reached in ~5.5 mo, an enrollment rate of 3.6 females per month. In contrast, males were enrolled at a rate of ~2.5 per month.

Retention throughout the study period was high, with 38 of 40 participants (95%) completing the intervention. All participants who completed the intervention also completed all scheduled assessments, resulting in complete outcome data among completers. During the 12-wk intervention period, 2 participants (5.0%) in the krill oil group withdrew from the study, resulting in 38 participants who completed the final assessment. One participant withdrew within 5 d of intervention initiation due to a migraine flare-up, and 1 participant withdrew for personal reasons. No participants were discontinued during the intervention, and no withdrawals were attributed to serious adverse events.

Adherence to the study supplement was high across both groups. Based on self-reported capsule intake (n = 37), mean adherence was 95.1% (SD = 5.8) in the krill oil group and 91.7% (SD = 12.0) among controls. Using capsule counts (n = 36), mean adherence was ~80.8% (SD = 9.9) in the krill oil group and 77.7% (SD = 12.1) among controls. When applying the prespecified adherence criterion of ≥70% daily capsule consumption, 100% of participants in the krill oil group (n = 18) and 89.5% of controls (n = 17) met this threshold based on self-report. Similarly, 83.3% of participants in the krill oil group (n = 15) and 83.3% of controls (n = 15) met the ≥70% adherence threshold based on capsule counts.

The intervention's acceptability, assessed with the Medication Acceptability Questionnaire, was generally high at the end of the study. Mean scores for convenience, taste, appearance, and tolerability ranged from ~71% to ~85%, indicating good overall acceptability of the study products. Perceived efficacy scores were lower and more variable (mean = 46.5%, SD = 24.6). Participants also reported being generally satisfied with the side effects, with a mean of 8.1 of 10 (SD = 2.6), and rated the overall acceptability at 7.7 of 10 (SD = 2.3). No substantial differences were observed between study groups (all *P* > 0.17).

Adverse events

In total, 30 participants (75%) reported ≥1 adverse event during the study, with 16 (80%) in the control group and 14 (70%) in the krill oil group. A total of 92 adverse events were recorded (controls: 58; the krill oil group: 34). No serious adverse events occurred in either group. Gastrointestinal symptoms were the most frequently reported adverse events, including diarrhea, constipation, flatulence, abdominal pain, and dyspepsia. Changes in appetite were also reported: decreased appetite by 6 participants (placebo: n = 5; krill oil: n = 1) and increased appetite by 3 participants (all in the control group). Unanticipated adverse events were more frequent in the control group (11 events) than in the krill oil group (3 events). Adverse events considered related to the intervention were also more frequent among controls (placebo: 5; krill oil: 2), all of which were gastrointestinal symptoms, including diarrhea, dyspepsia, flatulence, and abdominal pain. One participant in the krill oil group discontinued the intervention due to a migraine flare-up; no participants in the control group withdrew due to adverse events.

Biomarkers

The baseline ω -3 index (adjusted mean: 4.3%; 95% CI: 3.8%, 4.8%) was low among participants, consistent with limited dietary intake of ω -3 fatty acids. Changes in the ω -3 index are shown in Table 2 and Figure 2. The adjusted between-group mean difference in change of 3.0% (95% CI: 1.9, 4.0), corresponding to a very large effect (Cohen's $d = 4.40$). As expected, the ω -3 index increased significantly only in the krill oil group, with a marked increase evident by 6 wk (7.3%; 95% CI: 6.7%, 7.8%) and remaining stable through the final assessment (7.4%; 95% CI: 6.9%, 8.0%). In contrast, the placebo group showed no meaningful change over time (range: 4.5–4.7%). A similar pattern was seen with the ω -6: ω -3 ratio. No substantial changes in hs-CRP were observed.

Clinical pain outcomes

Across clinical pain outcomes, changes over time favored the krill oil group (Table 3). On the Graded Chronic Pain Scale–Revised average pain intensity item, the between-group difference in change was -1.05 points (95% CI: -2.72 , 0.62 ; Cohen's $d = -0.77$), corresponding to greater within-group pain intensity reductions in the krill oil group compared with placebo (-1.20 compared with -0.15 , respectively); shown in

Figure 3A. For pain interference, assessed using the PEG scale, the between-group difference in change was small (-0.09 ; 95% CI: -2.00 , 1.83 ; Cohen's $d = -0.06$), consistent with similar within-group changes in the krill oil and placebo groups (-0.76 compared with -0.68 , respectively).

Similar patterns were observed for WOMAC outcomes. The between-group difference in change was -5.0 for pain (95% CI: -18.9 , 9.0 ; Cohen's $d = -0.46$), -8.7 for stiffness (95% CI: -25.6 , 8.1 ; Cohen's $d = -0.66$), and -0.9 for physical function (95% CI: -13.1 , 11.2 ; Cohen's $d = -0.13$). For the WOMAC total score, the between-group difference in change was -2.4 (95% CI: -14.0 , 9.2 ; Cohen's $d = -0.32$). Within-group changes are presented in Table 3.

Beyond pain intensity and interference, participants receiving krill oil demonstrated a greater reduction in pain distribution, with a mean decrease of 3.5 total pain sites compared with 0.4 sites in the placebo group (Figure 3B). This corresponded to a 41% lower rate of pain sites at follow-up (rate ratio: 0.59; 95% CI: 0.41, 0.84).

Finally, consistent with these patterns, a greater proportion of participants in the krill oil group reported overall improvement on the PGIC compared with placebo (66.7% compared with 40.0%; adjusted odds ratio: 4.35; 95% CI: 0.97, 19.55).

TABLE 2
Biomarker outcomes at baseline and 12 wk, with between-group difference in change and effect size

Outcome	Krill Oil		Placebo		$\Delta KO - \Delta P$	95% CI	Cohen's d^2
	$t0^1$	$t2^1$	$t0^1$	$t2^1$			
ω -3 index (%)	4.29	7.41	4.49	4.67	2.95	1.89, 4.00	4.40
ω -6: ω -3 ratio	9.49	5.17	9.23	9.12	-4.22	-5.75 , -2.69	-4.19
log(hs-CRP)	0.79	0.68	0.74	0.84	-0.22	-1.05 , 0.62	-0.48

Abbreviations: CI, confidence interval; hs-CRP, high-sensitivity C-reactive protein; ΔKO , within-group change from baseline in the krill oil group; ΔP , within-group change from baseline in the placebo group.

¹ Values shown are model-based adjusted least-squares means at baseline and 12 wk from linear mixed-effects models, controlling for age, sex, and BMI.

² Cohen's d was calculated using the residual SD from the mixed-effects model ($\Delta\Delta$ standardized mean difference). Cohen's d interpreted as: small (~ 0.2), medium (~ 0.5), large (~ 0.8).

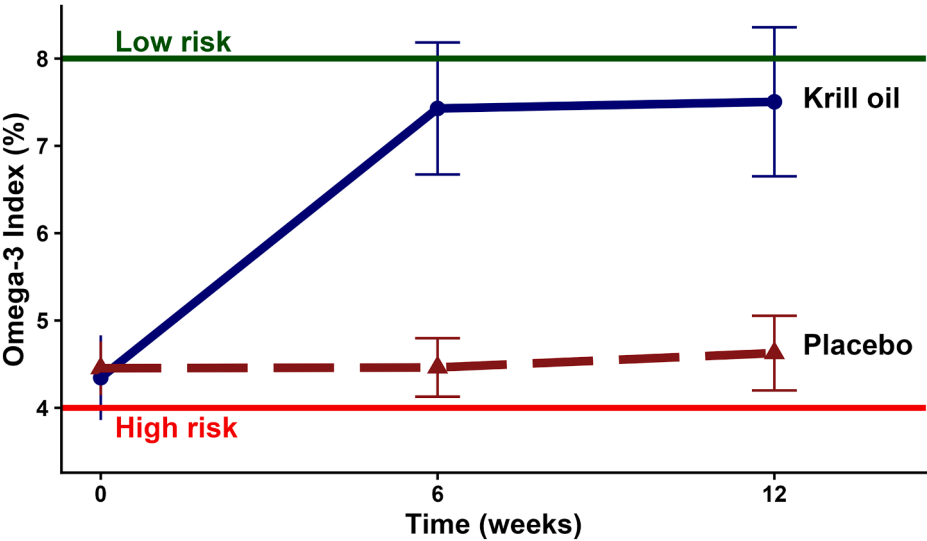


FIGURE 2. Change in ω -3 index over 12 wk in the KOPE pilot trial ($n = 40$). ω -3 Index (%EPA+DHA in erythrocytes) measured at baseline, 6 wk, and 12 wk in participants randomly assigned to krill oil or placebo. Horizontal reference lines denote ω -3 index thresholds: $\leq 4\%$ indicates high cardiovascular and mortality risk, whereas $\geq 8\%$ indicates low risk. KOPE, Krill Oil for Pain in Elders.

TABLE 3
Changes in clinical pain outcomes from baseline to 12 wk

Outcomes	Krill Oil		Placebo		Δ KO – Δ P	95% CI	Cohen's d ²
	t0 ¹	t2 ¹	t0 ¹	t2 ¹			
Pain intensity (0–10)	5.52	4.32	5.19	5.04	–1.05	–2.72, 0.62	–0.77
PEG score (0–10)	5.18	4.42	5.82	5.14	–0.09	–2.00, 1.83	–0.06
WOMAC pain (0–100)	44.96	35.18	45.13	40.32	–4.96	–18.87, 8.95	–0.46
WOMAC stiffness (0–100)	48.57	39.27	56.30	55.73	–8.73	–25.58, 8.12	–0.66
WOMAC physical function (0–100)	39.00	33.01	40.91	35.84	–0.93	–13.10, 11.24	–0.13
WOMAC total score (0–100)	41.04	34.01	43.07	38.43	–2.39	–13.98, 9.20	–0.32
Total pain sites (count)	7.91	4.46	7.54	7.19	Rate ratio	95% CI	—
					0.59	0.41, 0.84	

Abbreviations: CI, confidence interval; PEG, Pain, Enjoyment of Life, and General Activity; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index; Δ KO, within-group change from baseline in the krill oil group; Δ P, within-group change from baseline in the placebo group.

¹ Values shown are model-based adjusted least-squares means at baseline and 12 wk from linear mixed-effects models, controlling for age, sex, and BMI.

² Cohen's d was calculated using the residual SD from the mixed-effects model ($\Delta\Delta$ standardized mean difference). Cohen's d interpreted as: small ($\sim$ 0.2), medium ($\sim$ 0.5), large ($\sim$ 0.8).

Physical function outcomes

No meaningful between-group differences were observed across self-reported and performance-based physical function measures (Table 4).

Discussion

This pilot randomized controlled trial shows that a 12-wk protocol of 4 g/d krill oil supplementation is feasible and acceptable among older adults with moderate-to-severe chronic MSK pain. Although previous studies have shown promise for the analgesic effects of krill oil, most have focused on middle-aged adults, condition- and site-specific pain (i.e., knee OA), with mild-to-moderate symptom severity, and have been conducted outside the United States [25,48–52]. In contrast, we targeted older adults with chronic MSK pain and pain-related interference on physical function, a population with a greater burden of pain, comorbidities, and risk factors, and for whom non-pharmacological pain management strategies are particularly needed. Adherence exceeding the predetermined feasibility threshold, together with favorable acceptability, suggests that a 4 g/d dosing strategy is acceptable and implementable in this population.

The feasibility outcomes from this pilot inform and support the design of future fully powered randomized trials. Recruitment targets were met ahead of schedule, retention was high, adherence exceeded the prespecified thresholds, and overall acceptability of the intervention was favorable. We found that females were enrolled more rapidly than males, which led us to modify recruitment materials specifically for males. This suggests that diverse, targeted recruitment strategies may be needed to ensure balanced sex representation in a larger trial. Retention and adherence exceeded the prespecified criteria, supporting the feasibility of a longer intervention period and suggesting that a higher adherence threshold of $\geq$ 80% may be a reasonable benchmark.

The robust increase in the ω -3 index in the krill oil group confirms biological uptake and supports its use as an objective adherence biomarker. Mean ω -3 index values increased from $\sim$ 4% at baseline to $\sim$ 8% by follow-up, representing a clinically meaningful change from a higher-risk to a lower-risk range [34, 53]. This response is consistent with dose-response predictions by Walker et al. [54], which estimate that with a baseline ω -3 index $\sim$ 4%–5%, $\sim$ 1.2–1.4 g/d of EPA+DHA (as triglycerides) would be required to reach an ω -3 index of 8%. Although Walker's model does not account for phospholipid-bound ω -3 sources such as krill oil, the model-based prediction closely

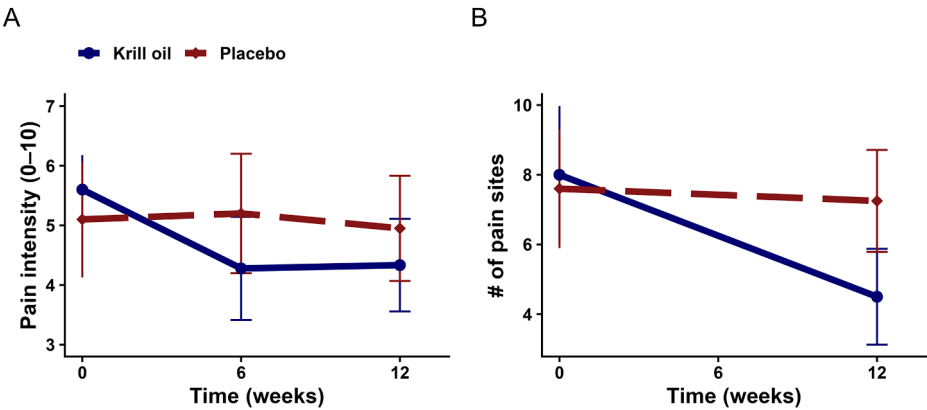


FIGURE 3. Trajectories of pain intensity and pain distribution over 12 wk in the Krill Oil for Pain in Elders (KOPE) pilot ($n = 40$). Mean pain intensity (panel A) and count of painful body sites (panel B) assessed at baseline, 6 wk, and 12 wk in the krill oil and placebo groups.

TABLE 4

Changes in physical function outcomes from baseline to 12 wk

Outcomes	Krill Oil		Placebo		$\Delta KO - \Delta P$	95% CI	Cohen's d ²
	t0 ¹	t2 ¹	t0 ¹	t2 ¹			
PAT-D total (0–76)	47.27	42.89	49.48	47.63	–2.53	–14.60, 9.55	–0.30
Mobility (0–28)	16.85	15.44	17.19	16.39	–0.61	–5.31, 4.10	–0.18
Basic ADLs (0–20)	14.51	13.10	15.48	14.96	–0.89	–5.21, 3.43	–0.27
Instrumental ADLs (0–28)	9.24	8.41	9.37	9.45	–0.92	–4.40, 2.57	–0.35
SPPB (0–12)	8.96	9.64	8.10	8.85	–0.072	–1.33, 1.19	–0.08
6-min walk distance (m)	375.1	362.9	358.5	353.9	–7.61	–76.5, 61.3	–0.24
Handgrip strength (kg)	30.82	32.20	29.22	30.76	–0.18	–6.93, 6.58	–0.06

Abbreviations: ADL, activities of daily living; CI, confidence interval; PAT-D, Pepper Assessment Tool for Disability; SPPB, Short Physical Performance Battery; ΔKO , within-group change from baseline in the krill oil group; ΔP , within-group change from baseline in the placebo group.

¹ Values shown are model-based adjusted least-squares means at baseline and 12 wk from linear mixed-effects models, controlling for age, sex, and BMI.

² Cohen's d was calculated using the residual SD from the mixed-effects model ($\Delta\Delta$ standardized mean difference). Cohen's d interpreted as: small (~ 0.2), medium (~ 0.5), large (~ 0.8).

aligns with the 1288 mg/d provided in this trial. Additionally, the model was derived from supplementation trials with a mean treatment duration of 13.6 wk, which is comparable to the 12-wk intervention period in the present study. Interestingly, the ω -3 index increased over 6 wk and stabilized thereafter, despite erythrocytes having a ~ 120 -d lifespan. It may be that incorporation into erythrocyte membranes may reach a steady state earlier than complete red blood cell turnover. Whether longer supplementation durations would lead to further increases in the ω -3 index warrants investigation with extended follow-up. On the contrary, habitual intake of fatty fish with this supplementation protocol may achieve the desired ω -3 index threshold of $\geq 8\%$ [55].

Although the study was not designed or powered to test efficacy, preliminary analyses of clinical pain outcomes consistently favored krill oil supplementation across multiple pain measures and domains. Participants receiving krill oil demonstrated larger reductions in average pain intensity, musculoskeletal pain and stiffness, pain distribution, and a greater likelihood of reporting overall improvement compared with placebo. These trends can be expected to continue with a longer supplementation period, as a meta-analysis of ω -3 supplementation randomized controlled trials indicated that benefits are noticeable within 1 mo and continue to improve by 6 mo [11]. These patterns suggest the potential efficacy of krill oil for more generalized, multisite, and high-impact pain phenotypes commonly seen in older adults. Reductions in multisite pain may be particularly relevant for older adults, as widespread pain is strongly associated with mobility limitations and disability. Importantly, effect size estimates are presented to aid interpretation and considerations for future trials, and should not be interpreted as evidence of efficacy.

On the contrary, krill oil supplementation did not result in meaningful improvements in self-reported or performance-based physical function over the 12-wk intervention period. As with pain interference, improvements in musculoskeletal health and physical function likely require longer intervention periods and sustained biological adaptation. ω -3 supplementation trials have shown limited and inconsistent evidence of efficacy for muscle mass, strength, and functional outcomes in older adults, particularly in interventions under 6 mo duration [56–59].

Although studies with krill oil are limited, a randomized controlled trial in generally healthy older adults showed that 4 g/d of krill oil for 6 mo significantly increased knee extensor strength, handgrip strength, and vastus lateralis muscle thickness, without improvements in SPPB performance [60,61]. In addition to intervention duration, improvements in physical function may be more difficult to achieve in older adults with high-impact chronic MSK pain.

Several limitations should be considered. As a pilot feasibility study with a relatively small sample size and short duration, it cannot inform the efficacy or safety of the intervention. The participants were recruited from a single academic center and surrounding areas and were mostly non-Hispanic White, which may limit generalizability to broader, more diverse populations. On the contrary, the participants in this study exhibit diet patterns and baseline ω -3 index levels that align with those in the general older population. Baseline dietary data confirmed low consumption of oily fish, with no participant reporting >1 serving per week (data not shown). Although adherence was high, as confirmed by self-report, capsule counts, and ω -3 index levels, long-term adherence may differ.

In conclusion, this pilot trial demonstrates that krill oil supplementation at 4 g/d is feasible and acceptable among older adults with chronic MSK pain. The feasibility and adherence data, together with consistent preliminary signals across pain outcomes, support and inform the design of a larger, fully powered randomized controlled trial. Given the burden of chronic pain in aging populations and the limitations of current pharmacologic pain management in older adults, krill oil may represent a promising, generally low risk, first-line or adjunctive non-pharmacological strategy to support pain management in aging. The findings support the practicality of implementing a krill oil nutritional intervention in this high-risk population.

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Author contributions

The authors' responsibilities were as follows – JAT, SDA, YC-A: designed the research; JAT, EC, KS: conducted the research; JAT, YC-A: performed statistical analyses; JAT, EC: wrote the first draft of the paper; KS, RP, SDA, YC-A: provided critical review; JAT: had primary responsibility for final content; and all authors: read and approved the final manuscript.

Declaration of AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) in order to assist with editing, clarity, and formatting of the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Conflict of interest

The authors report no conflicts of interest.

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Data availability

Data described in the manuscript, code book, and analytic code will be made available upon request, pending application and approval.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tjn.2026.101517>.

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